Habib University
Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Spring 2026
Instructor – Dr. Aamir Hasan - Section L4

Time = 20 Minutes **Quiz No 04** [20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

CLO Assessed – Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3) – *(applicable to both the questions.)*

Question 1 [10 points] [Probability Mass Function of a Random Variable]

We toss a fair coin 4 times. We define a Discrete Random Variable X as the number of consecutive Heads (we count a minimum two consecutive Heads) in 4 throws of the coin. Therefore, $\mathrm{X} = 0$ if the outcome of the 4-throw sequence is $\mathrm{H}, \mathrm{T}, \mathrm{H}, \mathrm{T}$ or $\mathrm{T}, \mathrm{T}, \mathrm{H}, \mathrm{T}$ and $\mathrm{X} = 3$ if the sequence is $\mathrm{T}, \mathrm{H}, \mathrm{H}, \mathrm{H}$.

Part a [5 points] Find and plot the probability mass function p.m.f for the DRV X.

Part b [5 points] Find the mean and variance of X.

Question 2 [10 points] [Probability Mass Function of a Random Variable]

We have a fair tetrahedral die with numbers 1, 2, 3 and 4. We roll the tetrahedral die n times and note the minimum value, m_n and the maximum value, M_n appearing during the n rolls. We define the Random Variable $D_n = M_n - m_n$ as the difference between the maximum and minimum values appearing in n rolls. As an example, for a three rolls sequence, if a particular outcome is 1, 3, 1, then the R.V, D_3 = 2 for that particular sequence.

Part a [6 points] Find and plot the probability mass function p.m.f for the DRV D_2. *(Note - this is for two throws of a tetrahedral die.)*

Part b [4 points] Just based on intuition, how do you think the distribution of Random Variable D_{16} behave?